

COLDPOD · DESIGN PRECIS

ColdPod design precis

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DOCUMENT CONTROL

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REVISION HISTORY

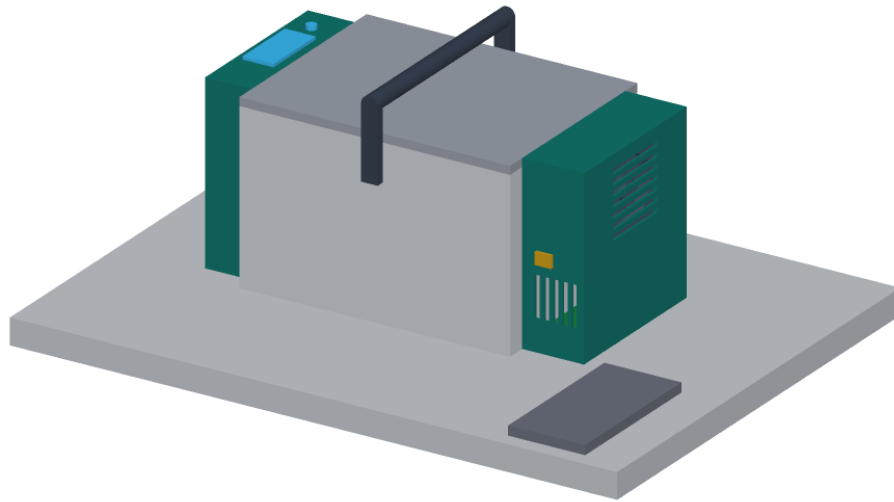
Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (CPD-DDR-001); numbers replaced by CPD-CAL-001; evaporator can, loop thermosiphon and TEC1-12703 class sized; smooth DC drive
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, design choices, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ColdPod design precis

ColdPod is a carry case about the size of a lunch cooler (368 x 214 x 207 mm) that holds 1.36 L of insulin or vaccines (24 pens) at 2 to 8 °C. The payload sits in an aluminium liner wrapped in a phase-change material (PCM) that melts at 5 °C, inside 25 mm vacuum-insulated panels. A Peltier module freezes the PCM from a 12 V socket, a solar panel or a USB-C charger, and a small LiFePO4 battery keeps it running on the road. A two-phase loop thermosiphon connects the two and carries heat in one direction only, so a stopped Peltier does not leak heat back in. A logger records the payload temperature every minute and raises alarms. The TRL 3 calculations (CPD-CAL-001) give about 13.0 h with no power at 43 °C, 26.5 h off-grid at 32 °C and 16.5 h off-grid at 43 °C, for \$295 in parts and 5.60 kg empty. The hold times meet their targets with thin margins; the design misses R12 (mass), R8 (the lid PCM pack cannot be refrozen by the Peltier) and R2 (a stuck-on driver can freeze the liner). Fixes are proposed, awaiting Amish.

ColdPod

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Grey: Table top, phone for scale

Figure 1. ColdPod on a table, with a phone for scale. Cooling head on the right end, battery and electronics bay with the e-paper display on the left end. Rendered from the parametric model `cad/src/model.py`.

■ How it works

1. **Charge the cold.** At a clinic, in a vehicle or under a solar panel, the Peltier module runs at up to 25 W and freezes the PCM jacket through the thermosiphon and the aluminium evaporator can around it. The same input charges

the battery at about 12 W. A melted jacket refreezes in about 5.5 h. The lid PCM pack has no cold path in this revision (see Open questions). The case works best if the payload goes in already cold from a refrigerator.

2. **Carry.** On the road, the controller runs the Peltier from the battery only as hard as needed to hold the evaporator can at about 2 °C, so the PCM stays frozen, and turns it off when the battery reaches 15 % so that the logger always has a reserve. A buck driver feeds the module smooth DC; on/off PWM would cost about 61 % more power at 43 °C.
3. **Hold passively.** When the battery is flat or the user switches cooling off, the PCM absorbs the heat that leaks through the walls. Because it melts at 5 °C, it cannot pull the payload below freezing the way water ice can.
4. **Watch.** A buffered probe among the payload, a liner probe, a cold-block probe and an ambient sensor are read every minute. The payload probe is logged with a timestamp on the controller's flash memory for 60 days or more.
5. **Alarm.** The display shows current, minimum and maximum temperature and time left. A buzzer, an LED and a phone notification warn after 10 min outside 2 to 8 °C, alarm at once at 0 °C or lower, and flag low battery, a failed sensor or a lid left open. The owner exports the log as a CSV file over Bluetooth Low Energy or USB. Nothing goes to the cloud.

ColdPod: heat flow, powered hold at 32 °C ambient (all values are estimates, CPD-CAL-001)

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Figure 2. Heat flow in powered hold at 32 °C ambient, can at 2 °C. All values are estimates from CPD-CAL-001: 0.091 W/K overall heat leak, TEC1-12703 class module at a coefficient of performance (COP) of 0.45, fan 0.6 W.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 4.

#	Component	Proposed choice	Notes
1	Outer shell	3D-printed PETG or ASA tub, 4 mm walls, 263 x 198 x 164 mm	Protects the VIPs from knocks and punctures
2	Lid	Printed cap with EPDM gasket and a 25 mm VIP plug	Latches closed; lid switch for the lid-open alarm
3	Bail handle and strap	Folding bail handle, padded shoulder strap	Rated 10 kg or more
4	Vacuum-insulated panels	Six panels, 25 mm, fumed silica core, ordered to size	VIPs cannot be cut or pierced; the heat-pipe crossing is a foam-filled gap between panels
5	PCM packs	About 0.97 kg of organic PCM melting at 5 °C (Rubitherm RT 5 HC class) in sealed HDPE pouches: a 0.69 kg jacket on four sides and the floor inside the evaporator can, and a 0.28 kg pack under the lid	Organic paraffins are stable for thousands of cycles (PATH PCM study)
6	Liner and rack	1.5 mm aluminium can, 175 x 110 x 80 mm outside (172 x 107 x 78.5 mm inside), with a removable printed rack for 24 pens in four layers of six	Spreads cold evenly around the payload
7	Evaporator can and thermosiphon	1.0 mm aluminium can lining the outer face of the PCM jacket (four sides and floor); two-phase loop thermosiphon: 8 mm copper evaporator loop around the base of the can, vapour riser and liquid return at the +X end, stainless sections through the wall, aluminium cold block 75 mm above the loop	One-way thermal link (design choices 2 and 3); sized in CPD-CAL-001, section G
8	Peltier module	40 x 40 mm, 127 couples, I _{max} about 3 A (TEC1-12703 class)	The TEC1-12706 cannot hold at 43 °C: its off-state conduction is too high (CPD-CAL-001, section E)
9	Heat sink and fan	Finned aluminium sink, 80 x 80 x 30 mm, 0.50 K/W or better with a 70 mm, 12 V fan and a finger guard	Rejects about 8.9 W at 32 °C and 19.7 W at 43 °C
10	End housings	Printed cooling head (+X end) and battery and electronics bay (−X end), vented	Keep the heat source and the battery outside the insulation
11	Battery	LiFePO ₄ , 4S1P 32700 cells, 12.8 V 6 Ah (76.8 Wh), BMS and fuse	Below the 100 Wh airline limit
12	Power board	USB-C PD sink (20 V), 12 V input with reverse-polarity protection, LiFePO ₄ charger, buck Peltier driver with an LC filter, fan driver, hardware cold-block cut-out	Off-the-shelf modules on perfboard; no custom PCB
13	Logger controller	nRF52840 module with real-time clock and 2 MB flash	Bluetooth Low Energy link to a phone
14	Temperature sensors	Glycol-buffered payload probe, liner probe, cold-block probe, ambient sensor	Digital sensors with ±0.5 °C or better accuracy
15	Display and alarm	2.13 in e-paper, buzzer, red and green LED	E-paper keeps the last reading visible with no power

Item 16 (wiring, fasteners, latches and consumables) is in the BOM but not modelled.

ColdPod: cutaway

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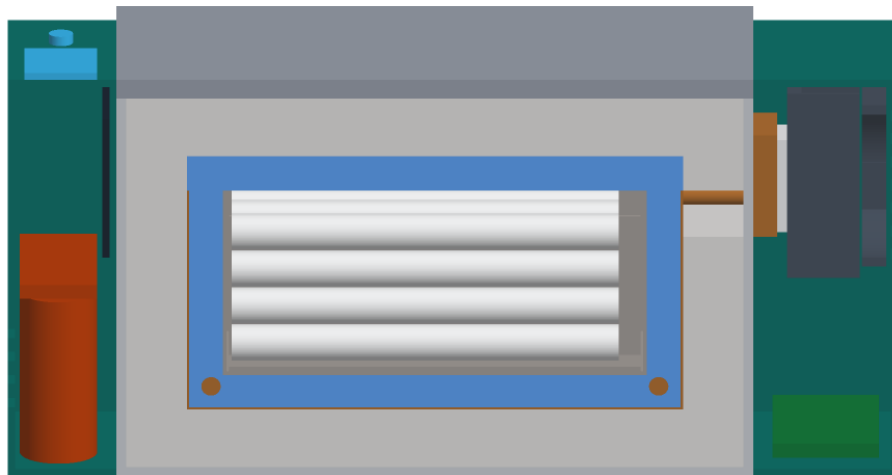


Figure 3. Section looking from the front. From the inside out: payload (white pens) in the rack and liner (6), PCM jacket and lid pack (5, blue) with the evaporator can and its loop (7, copper; the loop shows as two circles at the base), VIPs (4, light grey) and shell (1). The pipes (7) cross the right wall to the cold block, Peltier (8) and heat sink (9) in the cooling head. The battery (11) and logger (13) are in the left bay.

■ Numbers at TRL 3

All values come from the calculation note CPD-CAL-001 (docs/04-calcs/01-sizing.md), which states every assumption; the tags in brackets point to lines of its script, docs/04-calcs/sizing.py . They are paper estimates. The main inputs are an aged VIP conductivity of $0.007 \text{ W/(m}\cdot\text{K)}$ with explicit joint and edge losses, 180 kJ/kg of usable latent heat inside 2 to $8 \text{ }^{\circ}\text{C}$ for the PCM (Rubitherm RT range), a Peltier model derived from datasheet maxima (single-stage modules typically reach a COP of 0.3 to 0.7 , ThermalLM), and 65.3 Wh of the 76.8 Wh battery available for cooling.

Table 2. Hold times, charging, size, mass and cost.

Quantity	Value	Basis	Requirement
Payload	1.36 L; 24 insulin pens in four layers of six, up to 172 mm long	Liner less rack and probe [A2, A3]	R3 met
Heat leak	0.091 W/K: 1.82 W at 25 °C, 2.46 W at 32 °C, 3.46 W at 43 °C	Langmuir shape factor, joints, foam strip, pipes, wires [B9, B10]	
PCM	0.972 kg, 175 kJ (48.6 Wh) usable	Jacket 0.691 kg, lid pack 0.282 kg [C2, C3]	
Passive hold, PCM only	24.7 h at 25 °C, 18.3 h at 32 °C, 13.0 h at 43 °C	Jacket melts first [D1]	R4 at risk (10.0 h at +30 % leak)
Input power to hold	4.2 W at 25 °C, 7.5 W at 32 °C, 18.4 W at 43 °C	TEC1-12703 class, can at 2 °C [E3]	R7 at risk (lid pack)
Battery hold, then PCM	15.6 + 22.1 = 37.6 h at 25 °C; 8.7 + 17.8 = 26.5 h at 32 °C ; 3.5 + 13.0 = 16.5 h at 43 °C	65.3 Wh, then the PCM [F2]	R5 and R6 at risk
PCM refreeze from melted	5.5 h for the jacket at 25 °C; the lid pack has no cold path	Time-stepped model, 25 W to the module [G4]	R8 not met
Battery recharge	about 6.7 h at 12 W alongside the refreeze	45 W USB-C PD budget [G8]	
Heat rejected at the heat sink	5.2 W at 25 °C, 8.9 W at 32 °C, 19.7 W at 43 °C; sink base about 53 °C at 43 °C	Heat lifted plus module input [E6, E7]	Fan needed
Thermosiphon	Forward 2.50 W/K, reverse 0.0058 W/K; full function to 18° of tilt with the cooling head down	[H1, H5]	
Logger reserve	about 96 days	15 % of 76.8 Wh at a 5 mW allowance [J4]	R11 met
Log storage	1.38 MB for 60 days	1 record per minute, 16 bytes [J1]	R9 met on 2 MB flash
Size	368 x 214 x 207 mm (14.5 x 8.4 x 8.1 in) overall	Handle up [A6]	R13 met
Mass, empty	5.60 kg (12.3 lb)	Model volumes and densities [K2]; the evaporator can adds 0.25 kg	R12 not met (5.5 kg)
Parts cost	\$295	bom/bom.csv [L2]	R15 met (\$300)

■ Key design choices

Amish decided choices 1 to 9 on 2026-09-25, going with the recommendation in each case (CPD-DDR-001). The options are kept here for the record.

- 1. PCM at 5 °C instead of water ice (decided, D1).** Water ice stores about 334 kJ/kg, nearly twice as much, but a frozen ice pack sits at 0 °C or below and is a known cause of vaccine freezing ([Hanson et al., 2017](#)). A 5 °C PCM keeps the whole payload space inside the window. Options were (a) 5 °C organic PCM; (b) water with a conditioning step; (c) a 3 °C PCM. Decision: (a).

2. **A thermosiphon between the Peltier and the PCM (decided, D2).** A Peltier module that is switched off conducts heat well: 0.27 W/K for the chosen module, about three times the whole box's heat leak. Bolted straight to the liner, it would cut the passive hold at 43 °C to about 3.6 h. A gravity thermosiphon, with its condenser above its evaporator, only carries heat upward and out, so it acts as a thermal diode (forward to reverse ratio about 430). Options were (a) thermosiphon; (b) a mechanical disconnect that lifts the cold block away when power is lost; (c) direct mounting. Decision: (a), with (b) as the fallback if the thermosiphon cannot be made reliable at this size. TRL 3 sizes it as a two-phase loop.
3. **Evaporator on the outer face of the PCM, not on the liner (decided, D3).** Keeps the coldest surface (down to −2 °C during refreezing) separated from the payload by the PCM, which supports R2. At TRL 3 this is sized as an aluminium can lining the outer face of the jacket, because a small plate cannot freeze a paraffin jacket that wraps the liner.
4. **LiFePO₄, 76.8 Wh (decided, D4).** LiFePO₄ tolerates heat better than other lithium-ion chemistries and 76.8 Wh stays under the 100 Wh airline limit. The rejected 4S2P option (153.6 Wh, about \$30 more and 0.6 kg heavier) would give about 18.6 h at 43 °C off-grid.
5. **R6 redefined (decided, D5).** Accept about 16 h at 43 °C off-grid and run from a vehicle 12 V socket or a solar panel on long hot trips, rather than a larger battery or 35 mm VIPs with 1.6 kg of PCM (about 400 mm long and over 6 kg). Revisit after field data on real trip temperatures.
6. **Vacuum-insulated panels instead of foam (decided, D7).** Polyurethane foam of the same thickness (about 0.024 W/(m·K)) gives 0.160 W/K rather than 0.091 W/K, so the passive hold at 43 °C would drop from 13.0 h to about 7.4 h (CPD-CAL-001, P1 and P2), but it would save about \$45 and some mass. VIPs are used, with a foam version documented as a low-cost variant.
7. **Local alarms and Bluetooth Low Energy only (decided, D8).** No cellular or LoRa radio in the first build (either would add \$20 to \$40 and more power).
8. **Alarm thresholds (decided, D9).** Warn after 10 min outside 2 to 8 °C; alarm at once at 0 °C or lower on the payload probe; alarm if the liner probe reaches 1 °C (an early freeze warning); adjustable per product.
9. **Budget raised to \$300 (decided, D10).** The TRL 3 parts cost is \$295.

Sizing choices made at TRL 3, listed for Amish to confirm in docs/REVIEW.md : the evaporator can and loop thermosiphon (choices 2 and 3), the TEC1-12703 class module with smooth DC drive, and a 0.50 K/W heat sink.

ColdPod: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Outer shell
- 2 Lid cap with gasket
- 3 Bail handle and shoulder strap
- 4 Vacuum-insulated panels, 25 mm
- 5 PCM jacket and lid pack, 5 °C
- 6 Aluminium liner and payload rack
- 7 Evaporator can, loop thermosiphon, cold block
- 8 Peltier module, 40 x 40 mm
- 9 Hot-side heat sink and fan
- 10 End housings (cooling head, battery bay)
- 11 LiFePO4 pack, 12.8 V 6 Ah, with BMS
- 12 Power board (USB-C PD, 12 V, drivers)
- 13 Logger controller (nRF52840 class)
- 14 Temperature sensors (buffered probe)
- 15 E-paper display and alarm

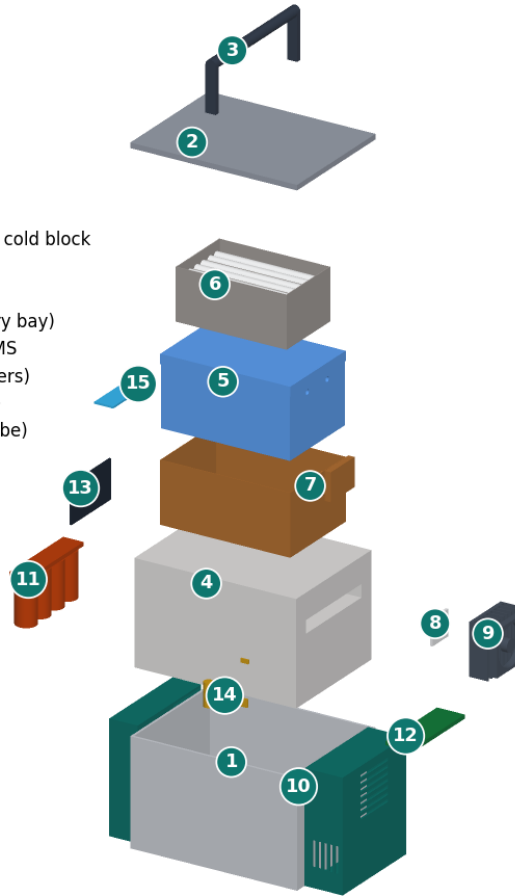


Figure 4. Exploded view with BOM numbers, from the parametric model. The payload pens rise with the liner (6); the evaporator can, loop and cold block (7) sit between the VIPs (4) and the PCM (5).

■ Safety

Safety: ColdPod is a research and educational prototype. It is not a medical device, is not WHO-prequalified and has not been cleared or approved by any regulator. Do not use it as the only protection for vaccines, insulin or other medicines. Follow national immunization program and manufacturer storage instructions, and use a qualified carrier and a calibrated logger for real products.

Safety: The LiFePO4 pack stores 76.8 Wh. Use it only with its BMS and fuse, charge it only between 0 and 45 °C (the controller must block charging outside that range), and never charge a damaged, swollen or wet pack. The battery sits in its own vented bay outside the insulation, away from the Peltier's hot side.

Safety: The PCM is a paraffin, which is combustible. It stays sealed in HDPE pouches, separated from the battery bay by the VIPs and the shell. Replace any leaking pouch; paraffin can also soften some plastics ([PATH PCM study](#)).

- **Hot surfaces.** At 43 °C ambient the heat sink base reaches about 53 °C (estimate, CPD-CAL-001, E7). It is enclosed by the vented cooling head, and the vents must not be blocked, covered or placed against skin or a bag.

- **Moving parts.** The fan has a finger guard inside the cooling head grille.
- **Electrical.** All voltages are 20 V DC or less. There are no mains-voltage parts in the box. Inputs are fused and protected against reverse polarity.
- **Freezing fault.** A stuck-on Peltier driver is the main way to freeze the payload. A hardware thermostat on the cold block cuts power to the Peltier below -5°C , independent of the firmware, and the liner and payload probes raise alarms. CPD-CAL-001 (section I) shows that this is not enough: once the PCM is frozen, the liner can follow the can down to about -4°C . A second hardware cut-out on the liner at 3°C , in series, is proposed, awaiting Amish.
- **Condensation.** Water condenses on the cold block and pipes. The cooling head drains outward and the electronics are conformal-coated.
- **VIP damage.** A punctured VIP loses most of its insulation value without any visible change. The shell protects the panels, and a rising Peltier duty cycle should trigger a service warning.
- **Lifting.** About 6.3 kg when loaded; carry with the strap across the body.

■ Open questions after TRL 3

- The lid PCM pack has no cold path to the evaporator, so the Peltier cannot refreeze it (R8), and in long powered holds at 43°C it melts and warms the top of the payload (R1, R7). Proposed, awaiting Amish: an aluminium plate under the lid pack that seats on the can rim when the lid closes; or no lid pack and a 20 mm jacket.
- Freeze fault (R2): add a hardware cut-out on the liner at 3°C . Proposed, awaiting Amish.
- Mass (R12): 5.60 kg against 5.5 kg. Thinner printed parts would reach about 5.39 kg. Proposed, awaiting Amish.
- Can the loop thermosiphon be made and charged reliably at this size, and does a motorbike carrier stay within about 18° of tilt with the cooling head down? If not, the mechanical disconnect (choice 2b) is the fallback.
- Confirm the off-state conductance and real COP of the chosen module with the chosen heat sink, and the sink's resistance with the fan.
- Confirm the VIP supplier, panel sizes, aged conductivity and edge losses; joints and edges are about 45 % of the heat leak.
- Confirm the usable latent heat of the PCM between 2 and 8°C , its supercooling and its behavior over many cycles.
- Where to place the payload probe so that it represents the warmest pen, not the average.
- Validate trip profiles, loads, alarm behavior and price with users through a partner (partner choice open, awaiting Amish).

Drawings and media: [general arrangement CPD-DWG-001](#), [concept blueprint sheet](#), [interactive 3D model](#).